

A practical strategy for valid partial prior-dependent possibilistic inference¹

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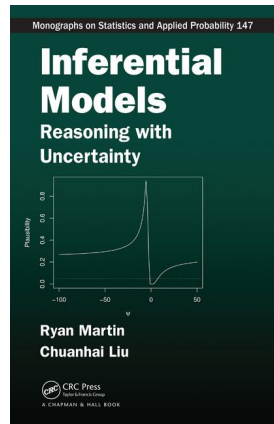
¹Joint work with Dominik Hose & Michael Hanss

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- Statistics aims to give reliable/valid uncertainty quantification about unknowns based on data, models, etc.
- Two dominant schools of thought:
 - *frequentist*
 - *Bayesian*
- Apparently neither is fully satisfactory
- *My goal:* Develop a “best of both worlds” alternative that
 - incorporates partial prior info when available
 - maintains reliability/validity
 - enjoys efficiency gains

- Relevant questions:
 - why partial prior info?
 - how to incorporate it?
 - what properties do we want?
- This talk:
 - IM background
 - partial priors & validity
 - *new construction*
 - computation



Statistical inference problem

- Two uncertain variables $(Y, \Theta) \in \mathbb{Y} \times \mathbb{T}$
 - Y represents observable data
 - Θ represents an unknown to be inferred
- Link is through a *statistical model*, a general imprecise probability pair $(\underline{P}_{Y,\Theta}, \overline{P}_{Y,\Theta})$
- Common scenario (my focus here):
 - precise conditional distribution $P_{Y|\Theta}$
 - imprecise/partial prior $\overline{P}_{\Theta}$
- Goal: learn from $Y = y$ about Θ :
 - uncertainty quantification
 - i.e., an imprecise “posterior distribution”

Definition.

An *inferential model* (IM) is a map from $(y, \dots)$ to a lower and upper probability pair on Θ :

$$\underline{\Pi}_y(A) = \text{degree of belief in } "\theta \in A," \text{ given } y$$

$$\overline{\Pi}_y(A) = 1 - \underline{\Pi}_y(A^c)$$

- Examples include: Bayes, fiducial, Dempster–Shafer,
- To ensure reliability, need statistical constraints

It is unacceptable if a procedure... of representing uncertain knowledge would, if used repeatedly, give systematically misleading conclusions (Reid & Cox 2015)

- Basic principle: if $\underline{\Pi}_Y(A)$ is large, infer A
- Idea: constrain $y \mapsto \underline{\Pi}_y(\cdot)$ so that

$\{\underline{\Pi}_Y(A) \text{ large, } \Theta \notin A\}$ has small $\overline{P}_{Y,\Theta}$ -probability

- For time's sake, jump directly to a stronger version...

Definition.

An IM $(y, \dots) \mapsto (\underline{\Pi}_y, \overline{\Pi}_y)$ is *strongly valid* (relative to $\overline{P}_{Y,\Theta}$) if

$$\overline{P}_{Y,\Theta}\{\pi_Y(\Theta) \leq \alpha\} \leq \alpha, \quad \text{for all } \alpha \in [0, 1]$$

where $\pi_y(\theta) = \overline{\Pi}(\{\theta\})$ is the plausibility contour

- Not immediately clear why it's called “strong”
- Structural consequences, e.g., *consonance*
- Statistical consequences, e.g.,
 - if $C_\alpha(y) := \{\theta : \pi_y(\theta) > \alpha\}$
 - then $\overline{P}_{Y,\Theta}\{C_\alpha(Y) \not\supseteq \Theta\} \leq \alpha$
- Vacuous prior case: “validity” $\implies$ “strong validity”

- <https://researchers.one/articles/21.01.00002>

An imprecise-probabilistic characterization of
frequentist statistical inference*

Ryan Martin[†]

- <https://researchers.one/articles/21.05.00001>

Valid and efficient imprecise-probabilistic inference
across a spectrum of partial prior information

Ryan Martin*

- Conference paper describes a strongly-valid IM construction, but doesn't fully justify the construction
- Forthcoming paper will give the full justification...
- *Key new idea*: outer consonant approximation
- More specifically:
 - likelihood-driven plausibility ordering on $\mathbb{Y} \times \mathbb{T}$
 - imprecise-probability-to-possibility transform
- Connection between IMs and possibility theory was made before,⁴ but now I'm doubling down on it

⁴<https://researchers.one/articles/20.08.00004>

Achieving strong validity, cont.

- Recall: $\bar{P}_{Y,\Theta}$ is determined by
 - precise $P_{Y|\Theta}$ with likelihood $p_\theta(y)$
 - imprecise $\bar{P}_\Theta$ with contour $q(\theta) = \bar{P}_\Theta(\{\theta\})$
- Define the IM $(\underline{P}_Y, \bar{P}_Y)$ to be consonant with contour

$$\pi_Y(\theta) = \bar{P}_{Y,\Theta}\{\eta(Y, \Theta) \leq \eta(y, \theta)\}$$

where η is the “relative likelihood”

$$\eta(y, \theta) = \frac{p_\theta(y) q(\theta)}{\sup_{\vartheta} \{p_\vartheta(y) q(\vartheta)\}}$$

- Incorporates partial prior info in *two places*
 - through q in the relative likelihood
 - through the $\bar{P}_{Y,\Theta}$ -probability calculation

- How to compute the $\bar{P}_{Y,\Theta}$ -probability that drives π_Y ?
- If $\bar{P}_\Theta$ is consonant, then the Choquet integral simplifies:

$$\pi_Y(\theta) = \int_0^1 \left[\sup_{\vartheta: q(\vartheta) > \beta} P_{Y|\vartheta} \{ \eta(Y, \vartheta) \leq \eta(y, \theta) \} \right] d\beta$$

- Suggests a simple/naive Monte Carlo strategy:

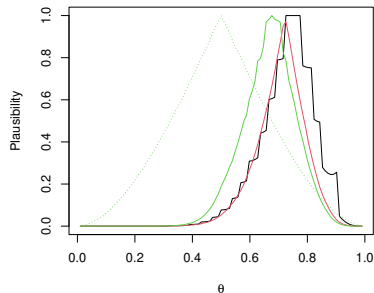
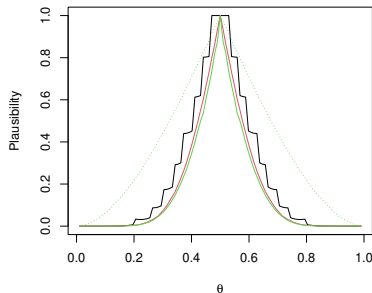
$$\pi_Y(\theta) \approx \int_0^1 \max_{s: q(\vartheta_s) > \beta} \left[\frac{1}{M} \sum_{m=1}^M 1 \{ \eta(Y_s^{(m)}, \vartheta_s) \leq \eta(y, \theta) \} \right] d\beta$$

where

- $\vartheta_1, \dots, \vartheta_S$ is a fixed, dense grid
- $Y_s^{(1)}, \dots, Y_s^{(M)} \stackrel{\text{iid}}{\sim} P_{Y|\vartheta_s}$, for $s = 1, \dots, S$

Illustration

- $P_{Y|\theta} = \text{Bin}(n, \theta)$
- Vacuous, **complete**, and **partial** prior info⁵
- Partial prior info is *more* influential than complete



⁵There are dimension-reduction steps taken that I'm not explaining...

- Strongly valid inference w/ partial priors is *possible*⁶
- New IM construction
 - likelihood-driven, prior accommodating
 - strongly valid (and “coherent”)
 - no “tuning parameters” for the user
- A simple computational strategy
- Can do lots more than I showed here
- I'm very excited about these new developments!

⁶Pun intended!

Thanks for your attention!

- Relevant papers:
 - <https://researchers.one/articles/21.05.00001>
 - this proceedings paper
- Exciting work in progress, paper(s) coming soon
- I'm teaching a PhD-level course now:



ST 790 (001) Fall 2022 Advanced Special Topics
Imprecise-Probabilistic Foundations of Statistics & Data Science

- Website: www4.stat.ncsu.edu/~rgmarti3